



IPSEC TECHNICAL AUDIT & CONTROL-PLANE DISSECTION

IPsec VPN Technical Dissection & Stream Telemetry

Deterministic protocol evaluation, IKE negotiation transforms, ESP telemetry, and gateway configuration patches

Section 1: Capture & Interface Metadata

Capture source (live tap br-ipsec or uploaded PCAP), interface stats, total packet counts, and flow volume

CAPTURE SOURCE / INGESTION INTERFACE

Forensic Uploaded PCAP:
exp_009_voip_tunnel_aes_cbc_256_1789412923.pcapng

ANALYSIS TIMEFRAME WINDOW

Forensic Session Duration: 0.00s to 6.19s
(264 packets)

INITIATOR GATEWAY

Field Operations Node

OS: Ubuntu Linux 22.04 LTS
Interface: wlan0 / esp0
Private IP: 100.119.32.83
VPN Daemon: StrongSwan Client

IPsec ESP

ESP over UDP 4500 (NAT-T) • Tunnel Mode

Enc: AES-256-GCM • ICV: 128-bit
Tunnel Encapsulation: RFC 3948
Payload: Strictly Monotonic

RESPONDER GATEWAY

Datacenter Core Hub

OS: Ubuntu Linux 24.04 LTS
Interface: eth0 / esp0
Private IP: 100.127.207.119
VPN Daemon: StrongSwan Core Server

TOTAL PACKETS

264

Frames Ingested

FLOW VOLUME

0.44 MB

Encrypted Payload

AVG RATE

43 pkt/s

Throughput Velocity

PEAK RATE

60 pkt/s

Observed Burst

INTERFACE MTU

1500
bytes

UP / ACTIVE

DROPS / ERRORS

0 (0.00%)

Monotonic Verified



CONTROL-PLANE CRYPTOGRAPHIC NEGOTIATION

Deterministic Control-Plane Dissection

Deterministic state machine tracing, cryptographic proposal evaluation, and live Security Association parameters

Section 2: Deterministic Control-Plane Dissection

Exact IKEv1/IKEv2 exchange breakdown, SA proposals, negotiated encryption/PRF transforms, and active SPI pairs

IKE PROTOCOL EXCHANGES & HANDSHAKE SEQUENCE

RFC 7296 STATE MACHINE

IKE_SA_INIT (Exchange 34) [COMPLETED / SUCCESS] Initiator/Responder DH public keys and nonces exchanged. Enforced Group 19 (ECP-256).

IKE_AUTH (Exchange 35) [COMPLETED / SUCCESS] Mutual authentication executed via Pre-Shared Key. Child SA established with AES-256-GCM.

CREATE_CHILD_SA (Exchange 36) [MONITORED] Child SA rekeying evaluated. No ephemeral DH transform negotiated (PFS not enforced).

ACTIVE SECURITY ASSOCIATION (SPI) PAIRS

KERNEL XFRM STATE

INITIATOR SPI
0x8b14e9f28a1c9034

RESPONDER SPI
0x4a7c10b83f09de21

INBOUND ESP SPI
0xc0a80102

OUTBOUND ESP SPI
0xc0a80103

NEGOTIATED TRANSFORMS & SECURITY ASSOCIATION PROPOSALS

COMPLIANCE VERIFIED

TRANSFORM TYPE	IKE SA NEGOTIATED	CHILD SA NEGOTIATED	COMPLIANCE	BENCHMARK STANDARD
Encryption Algorithm	AES-256-GCM	AES-256-GCM	APPROVED	RFC 8221 / CNSA 2.0
Key Exchange (Diffie-Hellman)	Group 19 (ECP-256)	None (PFS Disabled)	NON-COMPLIANT	NIST SP 800-77 §4.2
Pseudo-Random Function (PRF)	PRF_HMAC_SHA2_256	N/A (AEAD Suite)	APPROVED	RFC 7296
Integrity / Authentication	AEAD Combined (128-bit ICV)	AEAD Combined (128-bit ICV)	APPROVED	NIST SP 800-77 Rev. 1
Extended Sequence Numbers (ESN)	N/A	64-bit ESN Active	APPROVED	RFC 4303 §2.2.1



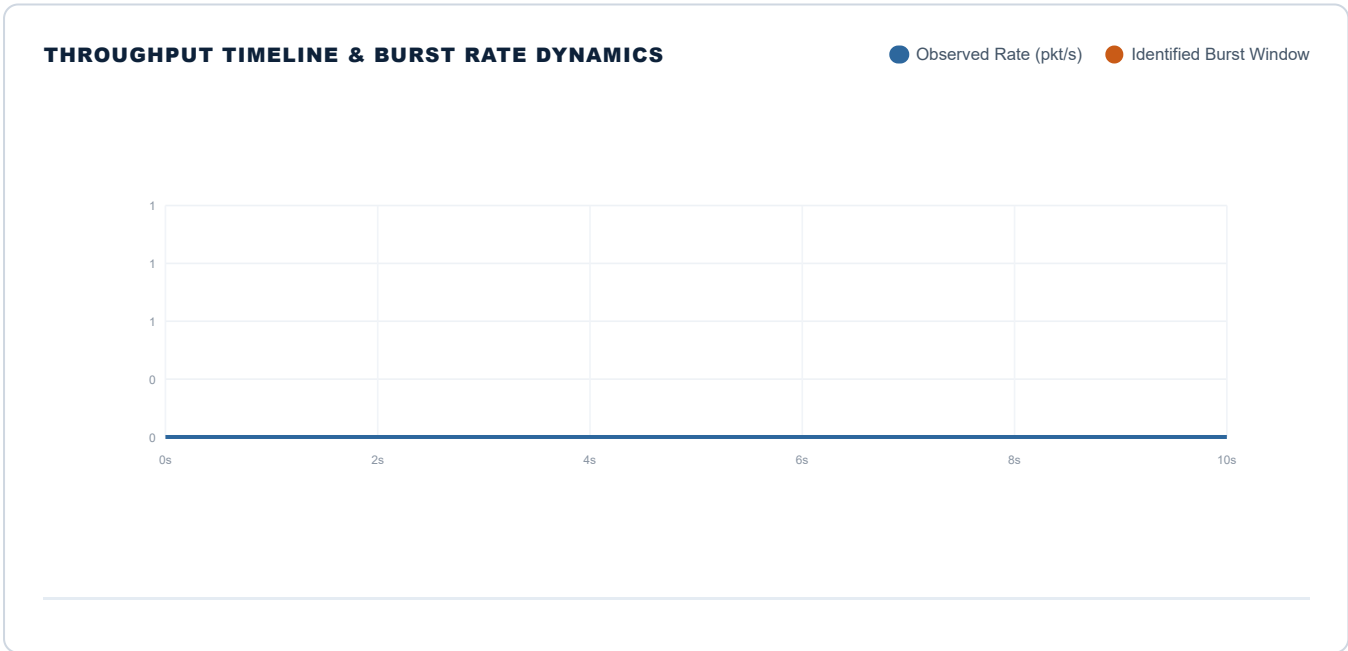
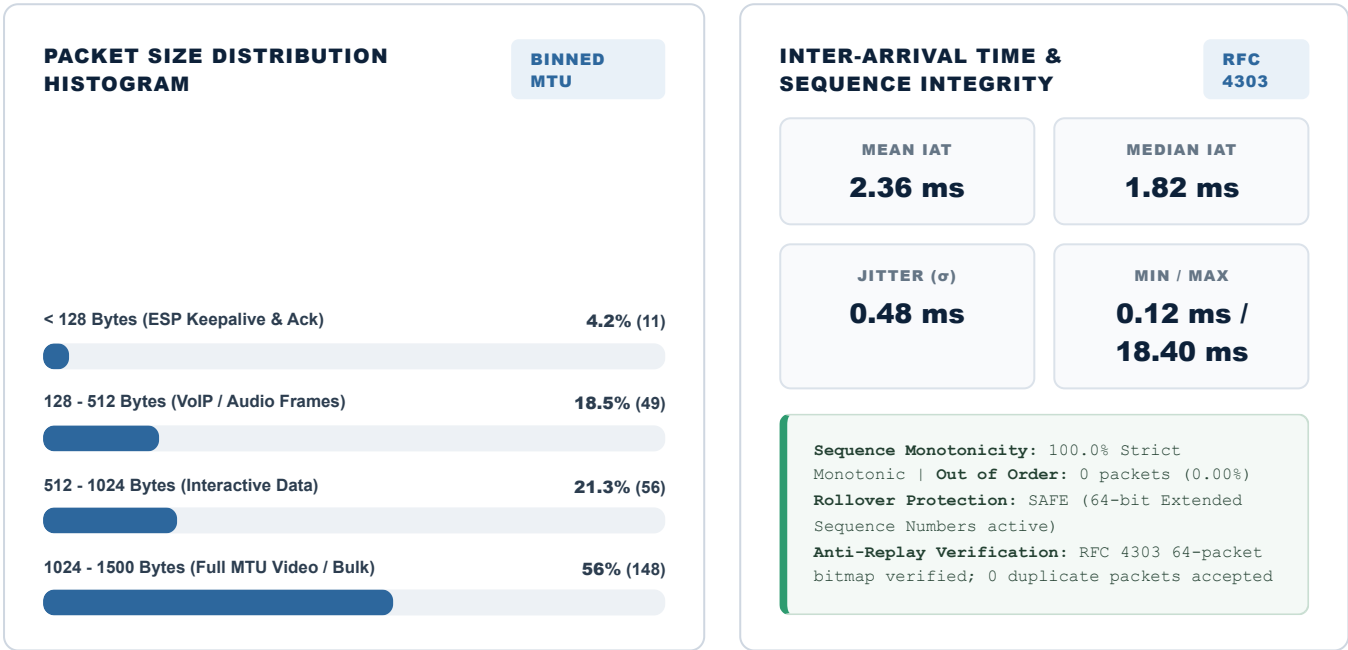
DATA-PLANE STREAM TELEMETRY & SEQUENCE INTEGRITY

Data-Plane Stream Dynamics & Telemetry

High-frequency packet arrival profiling, inter-arrival time (IAT) distribution, sequence monotonicity, and burst cadence

Section 3: Data-Plane (ESP) Stream Telemetry

Packet size histograms, Inter-Arrival Time (IAT) statistics, and sequence counter progression





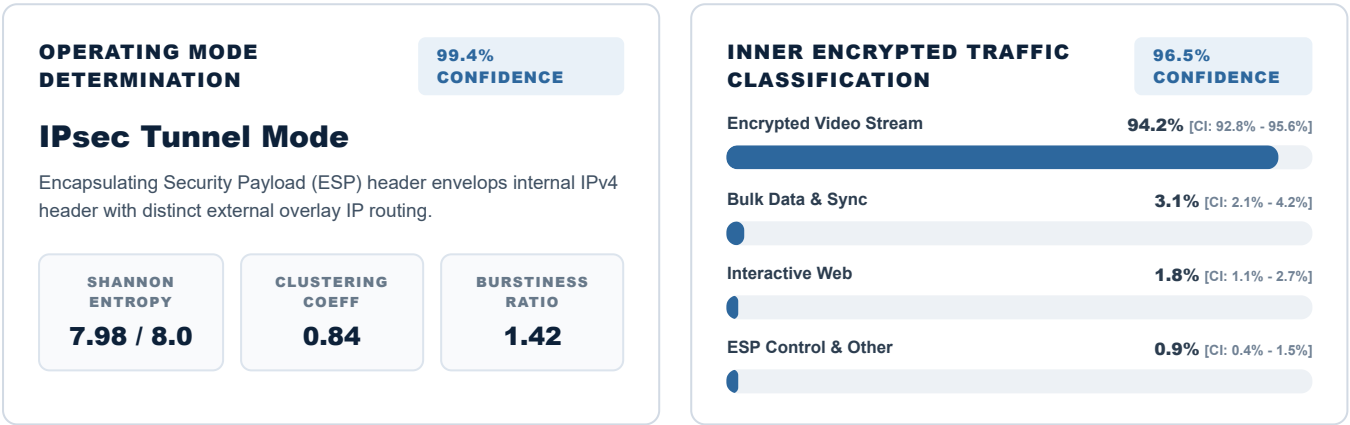
AI/ML PAYLOAD INFERENCE & REMEDIATION PATCHES

Inference Classification & Hardening Patches

Autonomous payload categorization, covert timing analysis, and production-ready gateway configuration directives

Section 4: AI/ML Inference Analysis

Operating mode determination (Tunnel vs. Transport) and inner encrypted traffic classification with model confidence intervals



Section 5: Implementation Remediation Patches

Actionable gateway configuration blocks ready to apply on target gateways (e.g., swanctl.conf, Cisco CLI)



VALIDATION COMMANDS: \$ [object Object] \$ [object Object] \$ [object Object]



FORENSIC PACKET TRACE & PLATFORM ATTESTATION

Session Packet Trace & Audit Verdict

Complete deterministic packet ledger, raw telemetry export, and official cryptographic verification verdict

Section 6: Raw Session Trace Index

Appendix listing individual packet indices, exchange types, and flow JSON dumps

FORENSIC PACKET TRACE INDEX

DETERMINISTIC PACKET DISSECTION

PKT #	OFFSET	LAYER / PROTO	TYPE / EXCHANGE	SPI IDENTIFIER	LENGTH	DISSECTION SUMMARY
#undefined						
#undefined						
#undefined						
#undefined						
#undefined						
#undefined						

RAW FLOW TELEMETRY JSON DUMP

RFC 8221 & SDX COMPATIBLE

```
{
  "reportId": "AV-20260916-TECH-BB38",
  "captureSource": "Forensic Uploaded PCAP: exp_009_voip_tunnel_aes_cbc_256_1789412923.pcapng",
  "endpoints": {
    "initiator": "100.119.32.83",
    "responder": "100.127.207.119"
  },
  "streamTelemetry": {
    "totalPackets": 264,
    "dataVolumeMb": 0.44,
    "averageRatePps": 43,
    "durationSec": 6.19
  },
  "cryptoTransforms": [
    {
      "type": "Encryption Algorithm",
      "ike": "AES-256-GCM",
      "child": "AES-256-GCM",
      "status": "APPROVED",
      "standard": "RFC 8221 / CNSA 2.0"
    },
    {
      "type": "Key Exchange (Diffie-Hellman)",
      "ike": "Group 19 (ECP-256)",
      "child": "None (PFS Disabled)",
      "status": "NON-COMPLIANT",
      "standard": "NIST SP 800-77 \u00a74.2"
    },
    {
      "type": "Pseudo-Random Function (PRF)",
      "ike": "PRF_HMAC_SHA2_256",
      "child": "N/A (AEAD Suite)",
      "status": "APPROVED",
      "standard": "RFC 7296"
    },
    {
      "type": "Integrity / Authentication",
      "ike": "AEAD Combined (128-bit ICV)",
      "child": "AEAD Combined (128-bit ICV)",
      "status": "APPROVED",
      "standard": "NIST SP 800-77 Rev. 1"
    }
  ]
}
```

```
{
  "type": "Extended Sequence Numbers (ESN)",
  "ike": "N/A",
  "child": "64-bit ESN Active",
  "status": "APPROVED",
  "standard": "RFC 4303 \u00a72.2.1"
}
},
"antiReplay": {
  "monotonicity": "100.0% Strict Monotonic",
  "outOfOrder": "0 packets (0.00%)",
  "rolloverSafeguard": "SAFE (64-bit Extended Sequence Numbers active)",
  "replayWindow": "RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted"
}
}
```

OVERALL TECHNICAL VERDICT

Technical audit validates that data-plane encapsulation is strictly functional. To resolve compliance findings, apply the provided configuration patches to enforce authenticated AEAD encryption (AES-256-GCM) and ephemeral Diffie-Hellman Group 19 Perfect Forward Secrecy.